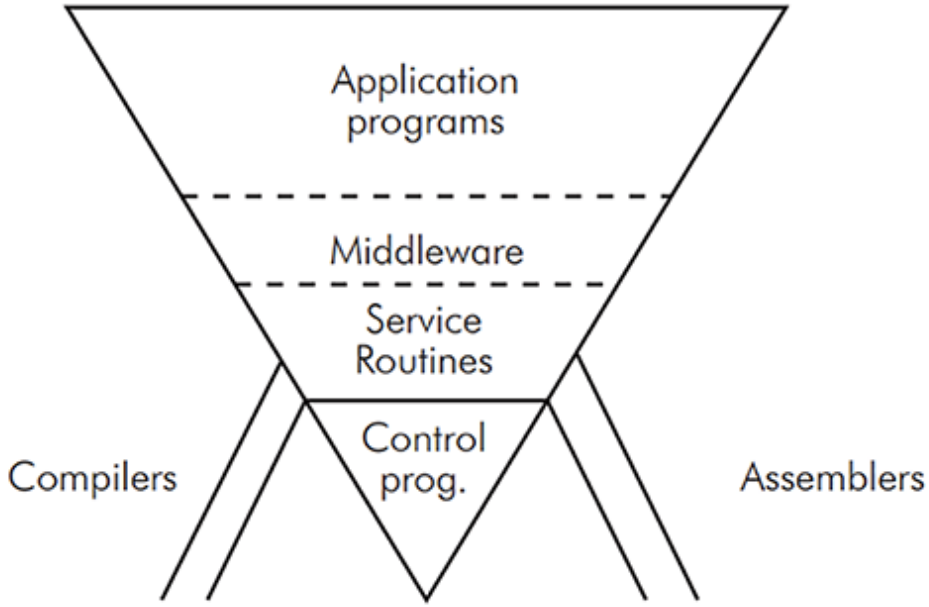


Figure 3. d'Agapeyeff's Inverted Pyramid

Figure B.2 Diagram from the 1968 NATO report (Source: NATO, http://mng.bz/ElJD)

It was not until the 1980s that middleware became mainstream. In the early 1980s, the telecommunications company AT&T developed Tuxedo, a platform based on the recently created middleware specification with messaging at its core. This architecture was called message-oriented middleware (MOM). The problem of integrating different applications or systems was already pervasive. During this decade, a group of universities in the United States built their own Computer Science Network (CSNET) because the recently created ARPANET was only accessible to the U.S. Department of Defense. These were some of the first steps toward greater integration. With the introduction of protocols such as DNS and TCP/IP on an international level, the internet was born, and numerous internet